affected by lower sales volume of cabinets, the divestiture of our Arrow and Moores businesses, and an unfavorable sales mix of international plumbing products, which, in aggregate, decreased sales by approximately two percent compared to 2016.

Net sales for 2016 were positively affected by increased sales volume of plumbing products, paints and other coating products and builders' hardware, which, in aggregate, increased sales by approximately five percent compared to 2015. Net sales for 2016 were also positively affected by favorable sales mix of cabinets and windows, and net selling price increases of North American windows and North American and international plumbing products, which, in aggregate, increased sales approximately one percent. Net sales for 2016 were negatively affected by lower sales volume of cabinets and lower net selling prices of paints and other coating products, which, in aggregate, decreased sales by approximately two percent.

Net sales for 2015 were positively affected by increased sales volume of plumbing products, paints and other coating products, windows and builders' hardware. Net sales for 2015 were also positively affected by net selling price increases of plumbing products, cabinets and windows, as well as sales mix of North American cabinets and windows. Net sales for 2015 were negatively affected by lower sales volume of cabinets and lower net selling prices of paints and other coating products.

Our gross profit margins were 34.2 percent, 33.4 percent and 31.5 percent in 2017, 2016 and 2015, respectively. The 2017 and 2016 gross profit margins were positively impacted by increased sales volume, a more favorable relationship between net selling prices and commodity costs, and cost savings initiatives. 2016 gross profit margins were negatively impacted by an increase in warranty costs resulting from a change in our estimate of expected future warranty claim costs.

Selling, general and administrative expenses as a percent of sales were 18.9 percent in 2017 compared with 19.1 percent in 2016 and 18.7 percent in 2015. Selling, general and administrative expenses as a percent of sales in 2017 reflect increased sales and the effect of cost containment measures, partially offset by an increase in strategic growth investments, stock-based compensation, health insurance costs and trade show costs. Selling, general and administrative expenses as a percent of sales in 2016 reflect strategic growth investments, ERP system implementation costs and higher insurance costs.

The following table reconciles reported operating profit to operating profit, as adjusted to exclude certain items, dollars in millions:

	2017	2016	2015
Operating profit, as reported	$ 1,169	$ 1,053	$ 914
Rationalization charges	4	22	18
Gain from sale of property and equipment	—	—	(5)
Operating profit, as adjusted	$ 1,173	$ 1,075	$ 927
Operating profit margins, as reported	15.3%	14.3%	12.8%
Operating profit margins, as adjusted	15.3%	14.6%	13.0%

Operating profit margins in 2017 and 2016 were positively affected by increased sales volume, cost savings initiatives, and a more favorable relationship between net selling prices and commodity costs. Operating profit margin in 2017 was negatively impacted by an increase in strategic growth investments and certain other expenses, including stock-based compensation, health insurance costs, trade show costs and increased head count. Operating profit margin in 2016 was negatively impacted by an increase in warranty costs by a business in our Windows and Other Specialty Products segment and an increase in strategic growth investments, as well as ERP system implementation costs and higher insurance costs.